

HUAZE (PATRICK) LIU

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EDUCATION

Harvey Mudd College

2022 – 2026

B.S. in Joint Computer Science & Mathematics (High Distinction)

Claremont, CA

- **Cumulative GPA:** 3.732 / 4.00 (**Major GPA:** 3.915 / 4.00)
- **Relevant Coursework:** Data Structures, Algorithms, Computer Vision, Neural Networks, Machine Learning, Reinforcement Learning, Multivariable Calculus, Linear Algebra, ODEs, Intermediate Probability, Real Analysis, Optimization, Advanced Systems Engineering, Digital Signal Processing, State Estimation.

PUBLICATIONS

[C1] Fay, K.*, Shukla, A.*, **Liu, H.**, Djapri, D., Buynitsky, A., Hansen, N. (2026). *Catch It All!: Generalizable Dynamic Catching with a Dexterous Hand*. 4th RSS Workshop on Dexterous Manipulation, in submission to ICRA 2027 (*equal contribution).

[C2] **Liu, H.***, Gao, Z.*, Mohanty, A. (2025). *Language-Driven Semantic Change Detection in Urban Maps via Multi-Modal Deep Learning*. The 38th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2025), Baltimore (*equal contribution).

[C3] **Liu, H.**, & Mohanty, A. (2025). *Conformal Prediction for Reliable Test-Time Uncertainty in Robotic Vision*. Southern California AI and Robotics Symposium (SCAIR 2025), Pomona.

RESEARCH EXPERIENCE

Georgia Institute of Technology

May 2026 – Present

Visiting Scholar, advised by Dr. Ye Zhao

Atlanta, GA

- Developing a whole-body tactile sensing system for the Unitree G1 humanoid to enable dense contact-rich perception.
- Designing a diffusion policy for navigation in unseen indoor environments using purely tactile sensing.

University of California, San Diego

June 2025 – March 2026

Visiting Summer Scholar, advised by Drs. Hao Su and Michael T. Tolley

La Jolla, CA

- Designed and trained a UniTree G1 upper-body interception policy under hardware torque and safety constraints. [C1]
- Implemented Sim-to-Real transfer using DAGger to deploy a vision-based controller at 50 Hz.

Harvey Mudd College

July 2024 – Dec. 2025

Undergraduate Research Intern, advised by Dr. Adyasha Mohanty

Claremont, CA

- Built a sensor-fusion map anomaly detection pipeline integrating image and LiDAR semantic segmentation. [C2]
- Evaluated split Conformal Prediction for segmentation and detection using vision foundation models. [C3]

Harvey Mudd College

Aug. 2023 – Dec. 2025

Undergraduate Research Intern, advised by Dr. Calden Wloka

Claremont, CA

- Built a video object-removal pipeline to automatically erase moving creatures from nature footage.
- Quantified the effects of padding-induced non-veridical representation on CNN performance with Mask R-CNN.

University of California, San Francisco

May – Aug. 2023

UCSF ci2 Summer Fellow, advised by Dr. Srikantan Nagarajan

San Francisco, CA

- Built Machine Learning models for Tinnitus classification using fMRI and MEG data, achieving 72.4% accuracy and benchmarking against Graph-CNN.

WORK EXPERIENCE

University of California, Berkeley

Aug. 2025 – May 2026

Full-stack Machine Learning Engineer Intern

Berkeley, CA

- Built a multi-pollutant Temporal Convolutional Network (TCN) pipeline in PyTorch for vehicle exhaust peak detection.
- Architected the data layer (TruckObject, DataWindow, checkpoint resolver) and an emission-factor calculator.
- Integrated the ML backend into a Flask web app with per-instrument model selection and cross-platform launchers.

Harvey Mudd College

Jan. 2024 – Dec. 2025

Teaching Assistant

Claremont, CA

- Tutored over 50 students in core concepts, provided homework assistance, guided projects, and graded assignments.
- Courses: CSCI153 – Computer Vision (2024 Fall, 2025 Spring, 2025 Fall), ENGR207 – Digital Signal Processing (2025 Spring), and CSCI070 – Data Structures & Program Development (2024 Spring, 2024 Fall).

PROJECTS

ProxWeld for Vulcan OmniPro 220 <i>TypeScript, Next.js</i> GitHub Repo	2026
• Built a manual-grounded AI support agent for the Vulcan OmniPro 220 welder using retrieval over bundled PDF manuals and the Claude Agent SDK. • Returned page-level citations, manual visuals, and structured wiring/troubleshooting helpers through a Next.js chat UI.	
Automated Video Object Removal Tool <i>Python</i> GitHub Repo	2025
• Architected a full video object removal pipeline integrating tracking, SAM2 masks, and diffusion-based inpainting. • Implemented temporal ghosting suppression post-process and improved video fidelity.	
Open-source Robotics Platforms CAD Collection <i>SolidWorks</i> GitHub Repo	2025
• Built SolidWorks models and assemblies aimed at enabling affordable, open-source robotic platforms.	
Vision-based Localization & Path Planning TurtleBot System <i>Python, MATLAB</i> Report	2025
• Developed a vision-based search TurtleBot 4 using EKF and UKF for state estimation and target tracking. • Used the A* algorithm to compute and follow an optimal path to the GPS coordinates.	
Multi-Objective Acquisition Function for Bayesian Optimization <i>Python</i>	2025
• Developed a lightweight multi-objective acquisition function benchmarked on SVM and CNN, reducing latency by 8% compared to EI and UCB.	
Vision Transformer for Traffic Object Detection <i>Python</i> GitHub Repo	2024
• Fine-tuned DETR on the KITTI dataset, achieving 71.0% mAP and 43.5% mAP (50:95) with 18.1 ms inference time.	
Dynamic Time Warping with Confidence Intervals <i>Python</i> GitHub Repo	2024
• Applied Dynamic Time Warping to develop new metrics and visualizations for audio alignment confidence.	
Spampede Game <i>Java</i> GitHub Repo	2023
• Designed a custom “Snake Game” UI and backend in Java using the MVC software design pattern; added an automated mode that finds optimal paths via BFS.	

SKILLS

Programming Languages: Python, C++, SQL, C#, Java, MATLAB, LaTeX, R
Frameworks / Libraries: PyTorch, NumPy, Pandas, OpenCV, Scikit-learn, Hugging Face, Flask
Software / Tools: Git, Linux, ROS/ROS2, Isaac Sim/Lab, MuJoCo, SolidWorks, Docker, Jupyter, Visual Studio
Technical Skills: 2D/3D Computer Vision, Deep Learning, Sensor Fusion, Reinforcement Learning, Robotics Control

AWARDS & SCHOLARSHIPS

Harvey Mudd College Dean’s List	2022 – 2026
Harvey S. Mudd Merit Award – Awarded \$10,000 annually for academic merit (\$40,000 total).	2022 – 2026
HMC Shanahan Projects Fund – Awarded \$11,000 for an underwater autonomous pool-cleaning system.	2024 – 2025
HMC Leeds Student Travel Grant – Awarded \$3,000 to attend the 38th ION GNSS+ Meeting, Baltimore, MD.	2025